Diagnostics and Event Management HLD

Synthetic AUTOSAR-inspired High-Level Design Knowledge Document

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Field	Value
Document ID	SYN-HLD-DG-001
Version	1.0
Status	Approved synthetic baseline
Classification	Synthetic academic test data
Intended use	RAG, OCR, extraction, comparison, and evaluation
Authority	Not valid for production design or vehicle release

1. Diagnostic architecture

The synthetic diagnostics domain collects application events, stores event state, and provides diagnostic snapshots to an external diagnostic transport adapter.

2. Components

ID	Component	Provides	Requires	Main responsibility
SWC-DG-001	DiagnosticEventManager	If_EventStatus	If_EngineState, If_LockStatus	Debounces and classifies events
SWC-DG-002	FreezeFrameCollector	If_FreezeFrame	If_EventStatus, If_VehicleSpeed	Captures contextual values
SWC-DG-003	DiagnosticDataRouter	If_DiagnosticData	If_EventStatus, If_FreezeFrame	Routes approved diagnostic records
SWC-DG-004	HealthMonitor	If_HealthStatus	If_TaskHeartbeat	Detects stale application tasks

3. Event catalogue

Event ID	Event name	Set condition	Clear condition	Snapshot fields
EVT-001	VehicleSpeedInvalid	Vehicle speed invalid for 500 ms	Valid for 2 s	VehicleSpeed_kph, MotionS
EVT-002	PedalChannelMismatch	Pedal channel delta exceeds 8% for 100 ms	Delta below 3% for 1 s	PedalPosition_pct, PedalVal
EVT-003	DoorStateTimeout	Door state absent for 1 s	Three valid updates	Door states, LockState
EVT-004	ThermalDeratingActive	MaximumTorque_Nm below 60% of request	Limit above 80% for 5 s	RequestedTorque_Nm, MaximumTorque_Nm

4. Dependency rules

FreezeFrameCollector shall only capture values explicitly listed in the event catalogue. DiagnosticDataRouter shall not create application relationships that are not present in approved HLD evidence.

5. Intentional orphan reference

Event EVT-005, BatteryVoltageLow, references signal BatteryVoltage_V. No component or interface providing BatteryVoltage_V is defined in this knowledge pack. This controlled defect is used for missing-dependency evaluation.

6. Audit fields

Field	Description
event_timestamp	UTC timestamp of the qualified event
event_status	PENDING, CONFIRMED, or CLEARED
source_component	Component that reported the condition
evidence_reference	Document, page, and section reference
review_status	UNREVIEWED, ACCEPTED, or REJECTED